CSA5108 Cryptography and Network security

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Exp 19 : Triple DES in CBC Mode
Code:

```
#include <stdio.h>
#include <string.h>
#include <openssl/evp.h>

int main(void) {
   unsigned char key[24] = "123456789012345678901234";
   unsigned char iv[8] = "12345678";
   unsigned char plaintext[1024], ciphertext[1040];
   int len, total = 0;

   EVP_CIPHER_CTX *ctx = EVP_CIPHER_CTX_new();

   printf("Enter plaintext: ");
   if (!fgets((char *)plaintext, sizeof(plaintext), stdin)) {
      EVP_CIPHER_CTX_free(ctx);
      return 1;
   }

   plaintext[strcspn((char *)plaintext, "\n")] = '\0';

   if (!ctx ||
      EVP_EncryptInit_ex(ctx, EVP_des_ede3_cbc(),
                  NULL, key, iv) != 1) {
      printf("Encryption initialization failed.\n");
      EVP_CIPHER_CTX_free(ctx);
      return 1;
   }

   if (EVP_EncryptUpdate(ctx, ciphertext, &len,
                  plaintext, strlen((char *)plaintext)) != 1) {
```

```
        printf("Encryption failed.\n");
        EVP_CIPHER_CTX_free(ctx);
        return 1;
    }

    total = len;

    if (EVP_EncryptFinal_ex(ctx, ciphertext + total, &len) != 1) {
        printf("Encryption finalization failed.\n");
        EVP_CIPHER_CTX_free(ctx);
        return 1;
    }

    total += len;

    printf("Ciphertext (hex): ");
    for (int i = 0; i < total; i++)
        printf("%02X", ciphertext[i]);

    printf("\n");

    EVP_CIPHER_CTX_free(ctx);
    return 0;
}
```

